

Puzzle sheet

Combined events: Tree diagrams

$\frac{4}{5}$	$\frac{1}{5}$	$\frac{1}{125}$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{12}{125}$	$\frac{61}{125}$	$\frac{2}{25}$	$\frac{1}{5}$	$\frac{3}{8}$	$\frac{2}{25}$

$\frac{3}{16}$	$\frac{12}{125}$	$\frac{1}{8}$			

$\frac{1}{5}$	$\frac{1}{25}$	$\frac{61}{125}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{2}{25}$

The answer to each question below has a corresponding letter. The numbers in the grid above are the answers to the questions. Use the tree diagrams to help you answer the questions. Then fill in the boxes above with the letters that match the answers, to discover a well-known saying.

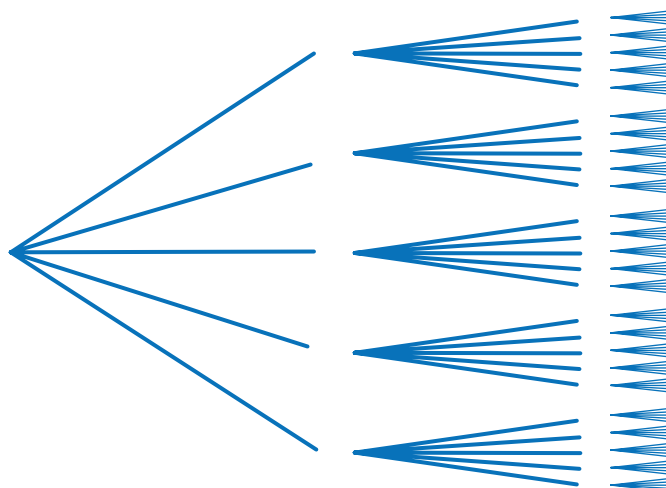
Questions

- 1 A test has three multiple-choice questions with responses labelled A, B, C, D and E. A student guesses the answer to each question. Illustrate the problem with a tree diagram and then find the probability that the student guesses:

a all the questions correctly _____ **M**

b only two questions correctly _____ **R**

c at least one question correctly. _____ **I**



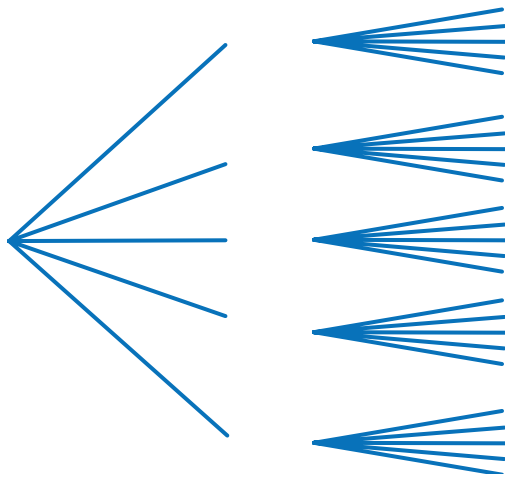
- 2 A pencil case contains five coloured pencils (red, blue, brown, pink and orange). A pencil is selected at random then it is returned before a second pencil is selected. Find the probability the pencils chosen are:

a blue then pink _____ **D**

b blue and pink in any order _____ **S**

c both the same colour _____ **O**

d different colours. _____ **C**



3 A die the shape of a tetrahedron has its faces numbered 1, 2, 3 and 4. The die is rolled twice and the number face down is recorded. Find the probability of rolling:

a the number 3 twice _____ **P**

b the number 2 on only one roll _____ **N**

c a 2 and a 3 _____ **E**

d two numbers the same _____ **U**

e a sum of 4. _____ **A**

